

Job Market Analysis: Web Scraping and SQL Analytics

Project Overview

This project demonstrates an end-to-end data analytics workflow, beginning with web scraping and ending with database analysis. Job listing data was collected from a public website using Python, cleaned and transformed using Pandas, stored in SQL Server, and analysed using SQL queries to identify hiring patterns and trends.

Objectives

- Collect job market data using web scraping techniques
- Clean and structure raw data for analysis
- Store data in a relational database
- Perform SQL-based exploratory analysis
- Demonstrate an end-to-end data analytics workflow

Tools and Technologies

- Python
- Pandas
- Requests
- BeautifulSoup
- SQL Server
- SQL Server Management Studio (SSMS)
- Microsoft Excel
- GitHub

Project Workflow

1. Data Collection

Developed a Python web scraper to extract job listing information from a publicly available website.

Data collected included:

- Job Title
- Company
- Location
- Application Link

2. Data Cleaning and Preparation

- Using Pandas, the scraped data was:
- Structured into a tabular format
- Checked for missing values
- Standardised and cleaned for consistency
- Exported for database integration

3. Database Development

Imported the cleaned dataset into SQL Server and created a dedicated database for analysis.

Database tasks included:

- Creating tables
- Importing cleaned data

Managing records
Preparing data for analysis

4. SQL Analysis

Performed exploratory analysis using SQL queries to investigate:

Total job postings

Hiring companies

Most common job titles

Geographic distribution of opportunities

Company-level hiring activity

Key Skills Demonstrated

Web Scraping

Data Collection

Data Cleaning

Data Transformation

Python Programming

SQL Querying

Database Management

Data Pipeline Development
